



Phase III Trial of Pegylated Liposomal Doxorubicin for Patients with Advanced and Refractory Desmoid Tumors

Huaiyuan Xu^{1,2}, Jinxin Hu^{1,2}, Yun Zhang^{2,3}, Aihua Zhuang^{2,4}, Yingchun Zhang^{2,5}, Jibin Li^{2,6}, Hao Wu^{1,2}, Guohui Song^{1,2}, Chuanquan Deng^{2,3}, Hongmin Chen^{1,2}, Weiqing Chen^{1,2}, Chuangzhong Deng^{1,2}, Anqi Wang^{1,2}, Huixiong Feng^{1,2}, Jinchang Lu^{1,2}, Qinglian Tang^{1,2}, Xiaojun Zhu^{1,2}, and Jin Wang^{1,2}

ABSTRACT

Purpose: To evaluate the efficacy and safety of pegylated liposomal doxorubicin (PLD) in the treatment of desmoid tumors (DT).

Patients and Methods: In this investigator-initiated, double-blind, phase III trial, we randomly assigned (in a 2:1 ratio) patients with advanced or refractory DTs to receive either PLD (50 mg/m² intravenously) or placebo every 4 weeks for six cycles. Crossover from placebo to PLD was permitted upon disease progression. The primary endpoint was progression-free survival (PFS). The secondary endpoints were objective response and safety.

Results: From November 2020 to March 2023, a total of 73 patients were assigned to receive PLD (49 patients) or placebo (24 patients). With a median follow-up of 16.1 months, PLD had a significantly longer PFS over placebo (not reached vs. 4.3 months),

with an HR of 0.05 (95% confidence interval, 0.01–0.17; $P < 0.001$). The 2-year PFS rates were 90.4% with the PLD group and 19.6% with placebo. PFS benefit favored PLD across most prespecified subgroups. The confirmed objective response was observed in 19 (40.4%) patients in the PLD group and in one (4.3%) in the placebo group ($P = 0.002$). Common grade 3 or higher adverse events with PLD included neutrophil count decreased (10.6%), mucositis oral (6.4%), and white cell decreased (4.3%). Significant between-group differences in certain patient-reported toxicities were observed ($P < 0.05$).

Conclusions: PLD significantly prolonged PFS and induced durable responses compared with placebo, with a favorable safety profile, for patients with progressive or symptomatic DTs.

See related commentary by Dossa et al., p. 2127

Introduction

Desmoid tumors (DT) are rare fibroblastic neoplasms, constituting less than 3% of all soft-tissue tumors, with an estimated annual incidence of three to five cases per million worldwide (1, 2). Although not malignant, DTs are often locally aggressive and may present with severe functional impairments, disfigurement, and pain, resulting in a substantial clinical burden (3, 4). Even after complete surgical resection, the risk of recurrence remains high, presenting a considerable challenge for disease management (5, 6).

The clinical course of DTs is highly variable and often unpredictable, which has led to considerable controversy about its optimal management strategy, especially for patients with progressive or highly symptomatic disease (7, 8). Innovative systemic treatments hold promise in providing more treatment strategies for this population; several regimens have already been investigated [e.g., tyrosine kinase inhibitors (TKI) and niraparacetat] or are currently under investigation [e.g., tegavivint (NCT03459469), AL102 (NCT04871282), and anlotinib (NCT05490667); refs. 9–11] but only niraparacetat has received FDA approval for the treatment of DTs based on the positive results of the DeFi trial. However, niraparacetat comes at the cost of limited tolerability [grade ≥ 3 adverse events (AE), 55%; discontinuation rate, 20%] and unusually high drug expenses, resulting in its inaccessibility for most patients with financial difficulties and in nonapproved regions. TKIs (e.g., sorafenib and pazopanib) have reported a modest response rate of 33% to 37%, but tolerability remains a challenge, with more than 20% of patients discontinuing treatment because of toxicities (9, 10). Consequently, there is a critical need for treatment regimens that improve survival and effectively manage toxicity, while maintaining affordability and ensuring broad accessibility.

Notably, the impressive impact on clinical outcomes in recurrent or progressive DTs was observed from conventional chemotherapeutic regimens. The methotrexate–vinblastine combination was the most frequently used nonanthracycline regimen, serving as the standard first-line regimen for children and demonstrating encouraging efficacy in adults (12–14). More surprisingly, anthracycline-containing regimens seem to be associated with a superior tumor response rate (33%–54%) and long-term survival benefits [median progression-free survival (PFS) of more than 20 months; refs. 14–16]. However, the associated toxicities, mainly cardiotoxicity and myelosuppression, must not be ignored behind the superior efficacy. Pegylated liposomal doxorubicin (PLD) is a modified anthracycline that reduces

¹Department of Musculoskeletal Oncology, Sun Yat-Sen University Cancer Center, Guangzhou, P. R. China. ²State Key Laboratory of Oncology in South China, Guangdong Provincial Clinical Research Center for Cancer, Sun Yat-sen University Cancer Center, Guangzhou, P. R. China. ³Department of Radiology, Sun Yat-sen University Cancer Center, Guangzhou, P. R. China. ⁴Department of Outpatient Clinics, Sun Yat-sen University Cancer Center, Guangzhou, P. R. China. ⁵Department of Pathology, Sun Yat-sen University Cancer Center, Guangzhou, P. R. China. ⁶Department of Clinical Research, Sun Yat-sen University Cancer Center, Guangzhou, P. R. China.

H. Xu, J. Hu, and Yun Zhang contributed equally to this article.

Clinical trial registration ID: NCT05561036

Corresponding Authors: Qinglian Tang, Department of Musculoskeletal Oncology, Sun Yat-Sen University Cancer Center, Guangzhou 510060, P. R. China; State Key Laboratory of Oncology in South China, Guangdong Provincial Clinical Research Center for Cancer, Sun Yat-sen University Cancer Center, Guangzhou 510060, P. R. China. E-mail: tangql@sysucc.org.cn; Xiaojun Zhu, zhuxj@sysucc.org.cn; and Jin Wang, wangjinbs@sysucc.org.cn

Clin Cancer Res 2026;32:2198–205

doi: 10.1158/1078-0432.CCR-25-3128

This open access article is distributed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license.

©2026 The Authors; Published by the American Association for Cancer Research

Translational Relevance

This is a placebo-controlled, double-blind, randomized, phase III trial that assesses the benefits and safety of pegylated liposomal doxorubicin (PLD) in the treatment of advanced and refractory desmoid tumors (DT). Our trial demonstrated that a six-cycle PLD regimen yields substantial progression-free survival benefits of 95.7% at 1 year and 90.4% at 2 years after randomization, with a 95.2% lower risk of disease progression over placebo in the overall population. The response benefits observed with PLD were also evident, with an encouraging objective response rate of 40.4%, along with durable responses (94.7% ongoing response as of data cutoff), in advanced DTs. PLD also had a favorable safety profile (grade ≥ 3 adverse events, 14.9%). Given its efficacy, safety, and accessibility, PLD could be considered a preferred treatment option with a favorable benefit/risk balance for patients with advanced DTs.

cardiotoxicity while preserving antitumor efficacy (17, 18). The efficacy and safety of PLD have been proven in various tumors, but its therapeutic value on DTs was only mentioned in a few small-sample retrospective studies, with robust evidence still lacking (19, 20). Based on this, we conducted a randomized, double-blind, placebo-controlled, phase III trial to evaluate the efficacy and safety of PLD in the treatment of DTs.

Patients and Methods

Study design and participants

In this investigator-initiated, double-blind, placebo-controlled, phase III trial, eligible patients aged 1 year or older with histopathology, IHC, molecular features, and musculoskeletal radiologist-confirmed DT were enrolled at a single center. Eligible patients were required to have an Eastern Cooperative Oncology Group performance status score ≤ 2 , at least one measurable lesion, radiographic progression ($\geq 10\%$ increase in maximum unidimensional measurement within 6 months before enrollment), and symptomatic disease with brief pain inventory score of ≥ 3 or unresectable disease or those with the possibility of postoperative disability. Key exclusion criteria included pregnant and nursing women, previous systemic therapy with PLD, currently receiving any other investigational drugs or anticancer therapy, active gastrointestinal bleeding, or severe cardiovascular diseases. Full eligibility criteria were listed in section 4 in the trial protocol.

The trial was conducted per the principles of Good Clinical Practice guidelines, the Declaration of Helsinki, and all applicable regulations, guidance, and local policies. Approval for the trial protocol was obtained from the responsible ethics committee. Written informed consent was obtained from each adult patient and from the parents or guardians of each pediatric patient.

Randomization and masking

Eligible patients were randomly assigned in a 2:1 ratio to receive either PLD or placebo. Randomization was done centrally by the statistical and data management center. A predefined randomization code was used to assign patients to study groups. Study participants, investigators, radiologists, and study site personnel remained masked to randomization codes until the end of treatment.

Treatment

Patients received 50 mg/m² PLD or placebo every 28-day cycle. Treatment was administered for up to six cycles from randomization or until disease progression or intolerable toxicity, whichever occurred first, as long as the patient had a benefit and did not meet other discontinuation criteria. Upon disease progression, patients were informed about their treatment, and those who received placebo were permitted to cross over to the PLD group if they still met the eligibility criteria. Dose reductions or interruptions were allowed to manage toxicity and are described in section 5 of the trial protocol. Patients could withdraw consent at any time for any reason or withdraw at the discretion of the investigator if untoward effects occurred. Patients could also be withdrawn for protocol violations and for administrative or safety reasons.

Endpoints and assessments

The primary endpoint was PFS as determined by investigators in accordance with Response Evaluation Criteria in Solid Tumors (RECIST), version 1.1, which was defined as the time from randomization to the first evidence of progression or death from any cause. The secondary endpoints were confirmed objective response, which was defined as complete response or partial response, and safety. Tumor response was assessed using MRI at baseline and every 3 months thereafter by at least two independent musculoskeletal radiologists, according to RECIST, version 1.1. Tumor volumes of each patient were also calculated from their three-dimensional models segmented through the ITK-SNAP program, and the longest diameter on the largest layer of the tumor displayed was measured in the image. Safety was assessed by monitoring vital signs and laboratory results throughout the treatment period until 30 days (90 days for patients who received PLD) after the end of the treatment. AEs were graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE), version 4.03. Treatment-related cardiotoxicity was monitored by echocardiographic surveillance. Exploratory endpoints were patient-reported outcomes (PRO) using CTCAE (PRO-CTCAE) version 1.0 before crossover. The assessment of PROs was performed at baseline and throughout the study optionally according to the patients' intention.

Statistical analysis

For the estimation of sample size, the median PFS was expected to be 15 months with PLD and 6 months with placebo based on the historical data (9). A total of 54 events were needed to provide a statistical power of 90% at two-sided α -error of 0.05 with 2:1 randomization. We determined a sample size of 72 patients, assuming a 12-month recruitment period, 24-month follow-up period, and 10% dropout.

Efficacy and safety were analyzed in all randomized patients who received at least one dose of the study drug. Time-to-event endpoints were estimated with the Kaplan-Meier method and compared using the log-rank test for between-group differences. HRs and associated 95% confidence intervals (CI) were assessed with a Cox proportional hazards model. For patients with loss to follow-up or without documented events, PFS was censored at the date of the last follow-up. The response rate with corresponding 95% CIs was calculated using the Clopper-Pearson method and compared using the χ^2 test. A hyperintense T2 score is defined as the percentage of tumor volume with hyperintense signal on T2-weighted images at baseline. Patients were dichotomized as high or low by the median of the hyperintense T2 scores (21). Favorable locations included lower extremity, intra-abdominal, and low back/pelvis/pelvic cavity

of the trunk (22). Full statistical analysis plan is available in section 10 of the trial protocol.

Results

Baseline characteristics

From November 12, 2020, to March 24, 2023, a total of 73 patients were enrolled and randomly assigned, of whom 70 received treatment (PLD, $n = 47$; placebo, $n = 23$). At the data cutoff (March 29, 2024), 49 patients completed protocol-specified treatment, 19 experienced disease progression, one patient was deceased, and one withdrew from the study because of substandard management (Fig. 1). Baseline characteristics were generally similar across trial groups and representative of the population of patients with DT (Table 1). Tumors were located in the lower extremities in 27 patients (37%; PLD vs. placebo: 24.5% vs. 62.5%) and in the head/neck region in seven patients (12.2% vs. 4.2%). The majority of patients presented with multifocal disease (75%; 71.4% vs. 83.3%). Overall, 76.7% of patients had previously undergone treatment that included systemic therapies, radiotherapy, or surgery. Representativeness of study participants is described in Supplementary Table S1.

Efficacy

As of cutoff, the overall median follow-up for PFS was 16.1 months, with 16.2 months (range, 1.0–37.1) and 13.1 months (range, 6.2–36.1) in the PLD and placebo groups, respectively. A total of 20 disease progression or death events occurred, including 19 disease progression (two with PLD and 17 with placebo) and one death (in the PLD group). The median PFS could not be estimated in the PLD group and was 4.3 months (95% CI, 3.3–5.2) in the placebo group. The estimated 1-year PFS rate was 95.7% (95% CI, 84%–98.9%) with PLD and 19.6% (95% CI, 5.5%–40%) with placebo, and the estimated 2-year PFS rates were 90.4% (95% CI, 70%–97.2%) and 19.6% (95% CI, 5.5%–40%), respectively. The risk of disease progression or death was 95.2% lower in the PLD group compared with the placebo group (HR, 0.05; 95% CI, 0.01–0.17; $P < 0.001$; Fig. 2A). PFS benefits favored PLD over placebo across most analyzed subgroups (Fig. 2B).

The confirmed objective response was observed in 19 patients (40.4%; 95% CI, 26.4%–55.7%) in the PLD group and in one (4.3%; 95% CI, 0.1%–21.9%) in the placebo group ($P = 0.002$) according to RECIST v1.1 (Supplementary Table S2). Patients in the PLD group

had four (8.5%) complete and 15 (31.9%) partial responses. Patients in the placebo group had one (4.3%) partial response. As per RECIST v1.1, the median best percentage change in the sum of the target lesions was –25.9% (range, –100.0% to 45.5%) with PLD and 8.1% (range, –37.6% to 51.0%) with placebo (Fig. 3A). Among the 47 patients treated with PLD, 42 patients (89.4%) had a reduction in target tumor size from baseline; eight (17.0%) had a reduction greater than 50% (Fig. 3A). With regard to the volume of the target tumor, the median best percent change was –59.6% (range, –98.6% to 48.6%) in the PLD group and 25.8% (range, –56.1% to 100.8%) in the placebo group (Supplementary Fig. S1). The time on treatment for patients with PLD and placebo is shown in Fig. 3B. As of the data cutoff, 18 of 19 (94.7%) responders in the PLD group were still having a response, with 13, eight, and one responding for more than 6 months, 1 year, and 2 years, respectively. The responder in the placebo group responded for 0.76 months. The earliest partial response to PLD was observed at 3.8 months, with a median time to response of 11.8 months (95% CI, 9.5–15.3). The median duration of response was not reached because of insufficient events. Examples of patients who achieved durable responses are shown in Supplementary Fig. S2.

Safety

Of 70 patients in the safety analysis set, the median duration of exposure was 5.9 months (range, 1.9–7.6) among the patients receiving PLD and 3 months (range, 0.9–6.8) among placebo. AEs of any grade occurred in 91.5% of patients with PLD and 36.4% with placebo (Table 2). Seven (14.9%) patients experienced grade 3 or 4 AEs in the PLD group, including neutrophil count decreased (10.6%), mucositis oral (6.4%), white cell decreased (4.3%), and palmar-plantar erythrodysesthesia syndrome (2.1%). No patients reported cardiotoxicity. No AEs led to the study interruption, but two (4.3%) patients reduced PLD because of AEs (one with myelosuppression and one with severe rash). Only one (2.1%) patient died as a result of pneumonia attributed to COVID-19 after one cycle of PLD treatment, which was deemed not related to PLD. No discontinuation due to AEs was observed in the two groups.

Details of toxicities reported by patients using PRO-CTCAE are provided in Supplementary Table S3. Of 11 measured, patient-reported PRO-CTCAE toxicities, six (appetite decreased, nausea, vomiting, rash, mouth or throat sores, and hand-foot syndrome) reached statistical significance between groups in comparing the

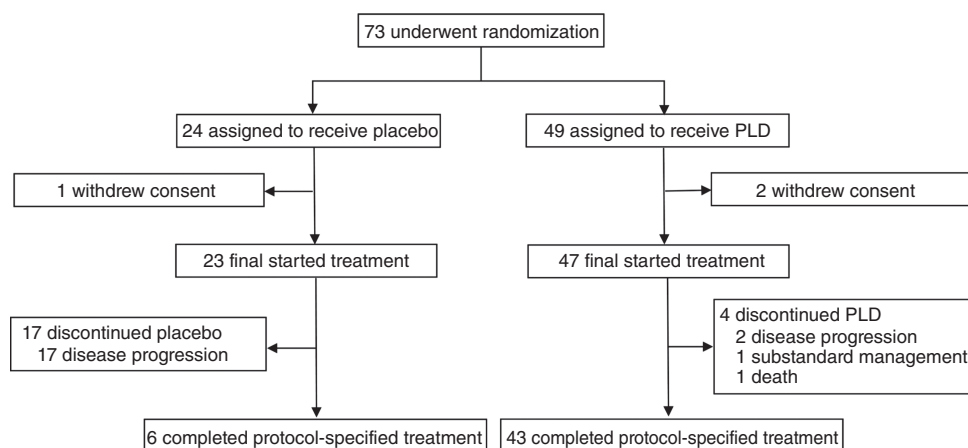


Figure 1.
Trial profile.

Table 1. Baseline characteristics in the intention-to-treat population^a.

Characteristic	Placebo (N = 24)	PLD (N = 49)
Median age, year (range)	27.5 (8–42)	26 (12–68)
<18 years	6 (25)	8 (16.3)
Sex, n (%)		
Female	15 (62.5)	33 (67.3)
Male	9 (37.5)	16 (32.7)
ECOG performance status, n (%)		
0	9 (37.5)	19 (38.8)
1	13 (54.2)	28 (57.1)
2	2 (8.3)	2 (4.1)
Hyperintense T2 scores ^b		
High	15 (62.5)	25 (51)
Low	9 (37.5)	24 (49)
Site of tumor, n (%)		
Lower extremity	15 (62.5)	12 (24.5)
Head/neck	1 (4.2)	6 (12.2)
Upper extremity	1 (4.2)	5 (10.2)
Intra-abdominal	1 (4.2)	4 (8.2)
Trunk	6 (25)	22 (44.9)
Favorable location ^c	18 (75)	26 (53.1)
Focal category, n (%)		
Single	4 (16.7)	14 (28.6)
Multifocal	20 (83.3)	35 (71.4)
Median sum of target lesions at randomization, cm (range)	15.7 (5.5–46.7)	15.9 (3–59.2)
Family history of FAP, n (%)	0	1 (2)
CTNNB1 mutation, n (%)		
S45 mutation	5 (20.8)	7 (14.3)
T41 mutation	9 (37.5)	11 (22.4)
Wild type	4 (16.7)	5 (10.2)
Unknown	6 (25)	26 (53.1)
Previous treatment, n (%)		
Surgical resection	21 (87.5)	32 (65.3)
Radiotherapy	3 (12.5)	4 (8.2)
Systemic therapy	4 (16.7)	11 (22.4)
Enrollment reason, n (%)		
Unresectable disease	20 (83.3)	43 (87.8)
Possibility of postoperative disability	3 (12.5)	4 (8.2)
Radiographic progression	1 (4.2)	1 (2)
Symptomatic disease with BPI score ≥ 3	0	1 (2)

Data are median (range) or n (%).

Abbreviations: BPI, brief pain inventory; ECOG, Eastern Cooperative Oncology Group; FAP, familial adenomatous polyposis.

^aThere were no significant differences between groups at a two-sided α level of 0.05. Percentages may not total 100 because of rounding.

^bHyperintense T2 score is defined as the percentage of tumor volume with hyperintense signal on T2-weighted images at baseline. Patients were dichotomized as high or low by the median of the hyperintense T2 scores (21).

^cFavorable location included lower extremity, intra-abdominal, and low back/pelvis/pelvic cavity of the trunk (22).

proportion of patients with score ≥ 1 with baseline adjustment ($P < 0.05$; Supplementary Fig. S3).

Discussion

To our knowledge, this trial represented the first prospective, randomized comparative data of PLD in patients with advanced and refractory DTs. Although median PFS was not reached at a median follow-up of 16.1 months, this study met the primary endpoint and demonstrated the significant improvement in PFS with PLD over placebo. The six-cycle PLD monotherapy resulted in an impressive PFS of 95.7% at 1 year and 90.4% at 2 years after randomization, with a 95.2% lower risk of disease progression over placebo in the overall population. The Kaplan–Meier curves of PFS separated immediately after randomization and remained consistently separated

throughout the evaluation period. PFS benefits favored PLD across most evaluated subgroups. The response benefits observed with PLD were also evident, with an encouraging objective response rate (ORR) of 40.4%, alongside the deep and durable responses in advanced DTs. With regard to safety results, PLD was acceptable without unexpected AEs.

Historical retrospective series of anthracycline-based regimens, including preferred PLD, have reported an ORR of 33% to 54% in a few cases with advanced aggressive DTs (7, 19, 20). Herein, clinical benefits of PLD were consistent with the previous efficacy data, providing confirmatory evidence for its efficacy in a large comparative study and reinforcing its status as the preferred chemotherapy for this rare tumor. Moreover, the magnitude of PFS benefit from PLD monotherapy was considerable with an HR of 0.05 (95% CI, 0.01–0.17), which seems to be very promising based on the recent results in this population (9, 11). Considering all the caveats of

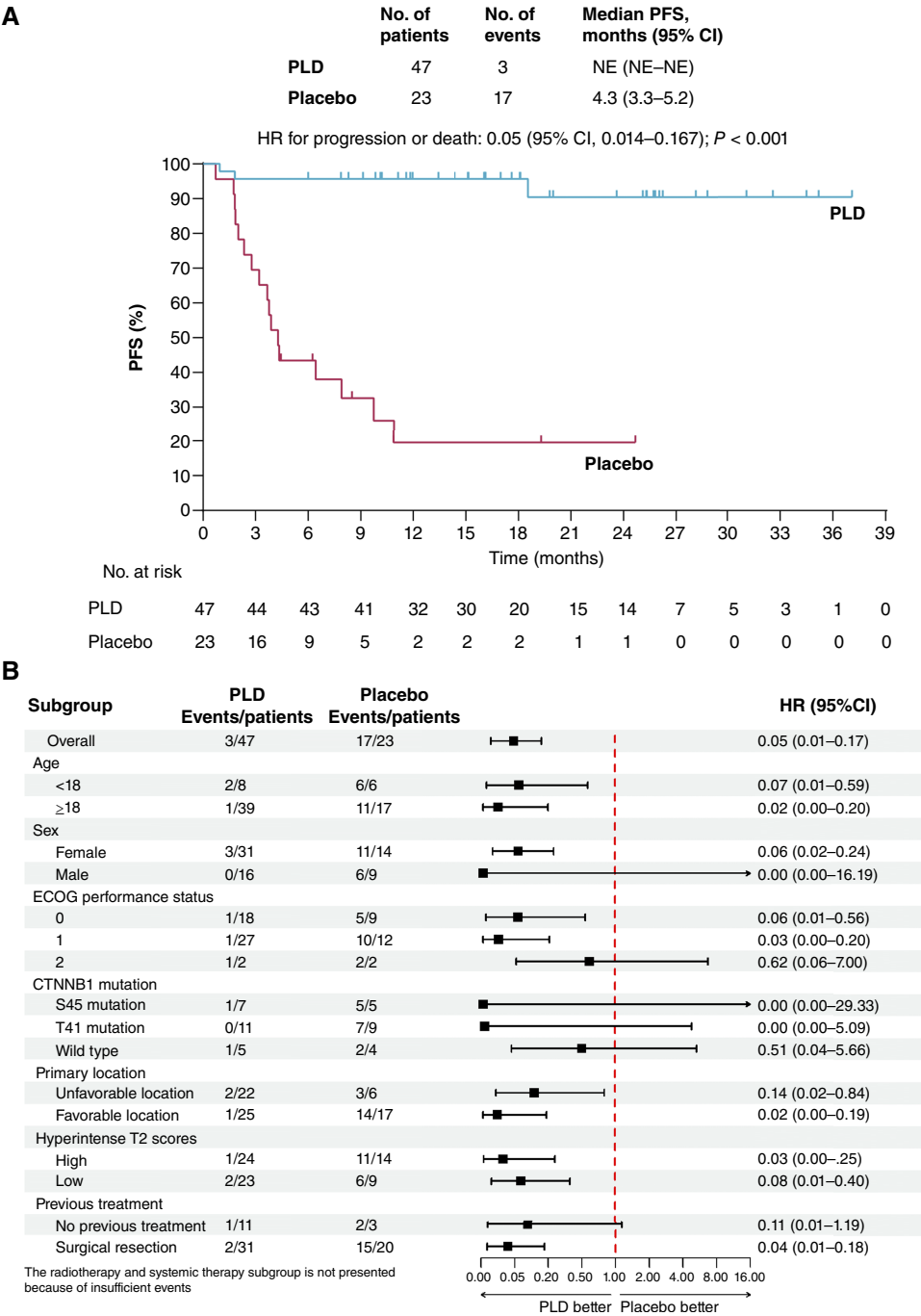


Figure 2. PFS and subgroup analysis. **A**, Kaplan–Meier estimates of PFS. Disease progression was identified by the investigators and included imaging-based progression according to RECIST 1.1 and clinical progression. Tick marks indicate censored data. **B**, Forest plot of PFS in prespecified subgroups. Disease progression was identified by the investigators and included imaging-based progression according to RECIST 1.1 and clinical progression. NE, not evaluated.

cross-trial comparisons, we have to admit the fact that the median PFS (4.3 months) in the placebo group was less than the expected median of 6 months and those reported in earlier trials, which may lead to the overestimation of actual benefit. Nevertheless, the well-balanced population in the two groups could alleviate this concern to some extent.

The shorter PFS observed in the placebo group may be primarily attributed to the inclusion of a higher proportion of patients with adverse prognostic factors. Tumor size, extra-abdominal location,

and positive surgical margins have been reported as independent predictors of disease progression in DTs (23). In our study, the enrolled population exhibited more unfavorable baseline features compared with previous trials, including a larger median tumor size (15.7 cm vs. 7.6–11.2 cm), a greater proportion of extra-abdominal lesions (95.8% vs. 49%–75%), and a higher rate of prior surgical resection (87.5% vs. 49%–61%; refs. 9, 11). These imbalances in baseline characteristics may have led to faster disease progression in the placebo group, thus contributing to the shorter PFS observed.

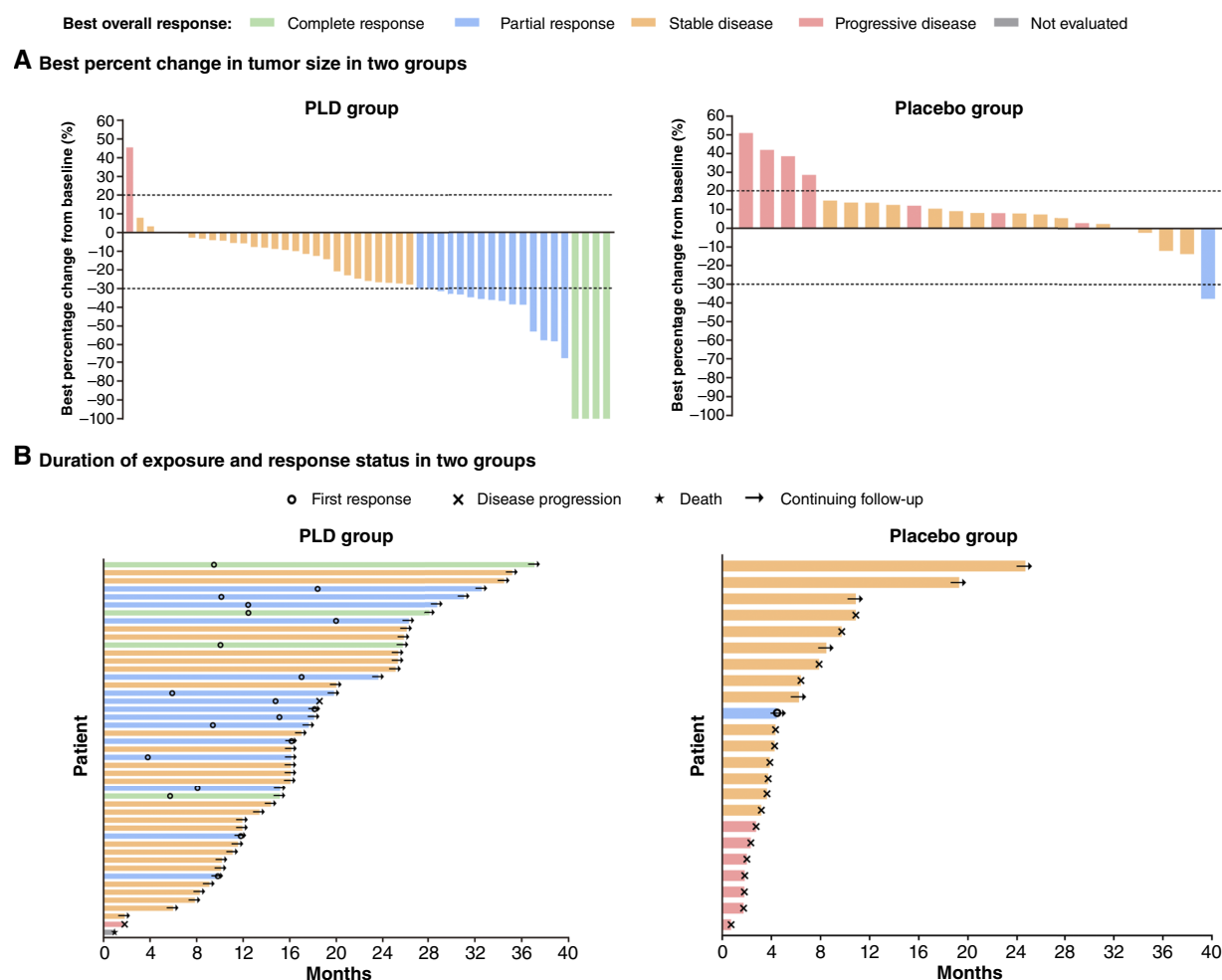


Figure 3.

Tumor responses and clinical outcomes. **A**, Waterfall plots of the percentage change in tumor size from baseline, as assessed by RECIST v1.1. Each bar represents a patient, with dashed lines indicating partial response (30% decrease) and progressive disease (20% increase). **B**, Swimmer plots of the duration of response and clinical outcomes in each group.

Additionally, the rapid progression in the placebo group resulted in a lack of objective responses (4.3%), lower than the historically reported response rates of 8% to 20% (9, 11). This discrepancy may be attributed to our small sample size, relatively brief follow-up period, and the unfavorable prognosis of the patient population.

Despite the differing baseline characteristics and the higher proportion of poor-prognosis patients in our trial, our results about the PFS and ORR were comparable with the previously reported efficacy of the individual drugs (1-year PFS, 85%–89%; 2-year PFS, 67%–81%; and ORR, 33%–41%; refs. 9–11) although comparisons between trials should be interpreted with caution. Given the FDA approval of nirogacestat and our equally promising results, we are optimistic that PLD may represent the next step toward improving outcomes for advanced DTs, in particular when considering the drug accessibility and economic cost. Maximizing survival and minimizing toxicity should be an important consideration in the treatment decision for patients with DTs with long life expectancy (24). As expected, the PLD treatment may not be associated with a discontinuation rate because of toxicity compared with other

existing agents (8%–20%; refs. 9–11). Similarly, the incidence of grade 3 or worse AEs followed a consistent pattern when compared with our trial with other studies (14.9% vs. 49%–56%; refs. 9–11), supporting the potentially preferable safety of PLD in the treatment of DTs. An important possible reason for this preferred safety is the fixed treatment duration and longer dose intervals of PLD (a total of six 28-day cycles), whereas other agents were continuously administered until disease progression or unacceptable toxicity. Importantly, the intermittent administration and PLD discontinuation did not affect the sustained response among responders in our study.

As a major safety concern about anthracyclines, no cardiotoxicity occurred owing to the structural advantage of PLD, which was consistent with previous experience (20). As previously reported (25, 26), the low risk of cumulative-dose cardiotoxicity may allow the long-term administration of PLD and offer long-term survival for some patients. However, this speculation remains to be proven formally. We also noted that two female responders had successful pregnancies and deliveries after discontinuing PLD, indicating that PLD may effectively maintain antitumor activity even under the

Table 2. Incidence of AEs according to trial group and grade^a.

Event ^b	PLD (N = 47)		Placebo (N = 23)	
	Grade 1 or 2	Grade 3 or 4	Grade 1 or 2	Grade 3 or 4
Mucositis oral	28 (60)	3 (6)	4 (18)	0
Palmar-plantar erythrodysesthesia syndrome	23 (49)	1 (2)	1 (5)	0
Neutrophil count decreased	15 (32)	5 (11)	0	0
Alopecia	18 (38)	0	0	0
Nausea	16 (34)	0	2 (9)	0
White cell decreased	13 (28)	2 (4)	0	0
Anorexia	14 (30)	0	0	0
Vomiting	10 (21)	0	0	0
Skin hyperpigmentation	9 (19)	0	0	0
Fatigue	7 (15)	0	3 (14)	0
Menstrual disorders	6 (13)	0	0	0
Constipation	5 (11)	0	0	0
Abdominal pain	5 (11)	0	1 (5)	0
Diarrhea	5 (11)	0	0	0
Elevated alanine aminotransferase	2 (4)	0	0	0
Respiratory symptoms	2 (4)	0	0	0
Anemia	2 (4)	0	1 (5)	0
Blurred vision	2 (4)	0	0	0
Flatulence	1 (2)	0	0	0
Sore throat	1 (2)	0	1 (5)	0
Elevated alkaline phosphatase	1 (2)	0	0	0
Urethral pain	1 (2)	0	0	0
Rash	1 (2)	0	0	0
Papulopustular rash	1 (2)	0	0	0
Insomnia	1 (2)	0	0	0
Decreased body weight	1 (2)	0	0	0
Complete atrioventricular block	1 (2)	0	0	0
Elevated lactate dehydrogenase	1 (2)	0	0	0
Gingival pain	1 (2)	0	0	0
Leukocytosis	0	0	1 (5)	0

Data are expressed as *n* (%).

^aThe safety population included all randomized patients who received at least one dose of the study drug.

^bAEs were graded according to the CTCAE, 4.03. Multiple occurrences of the same events were counted once per patient, according to the highest grade.

presence of high-risk factors, with the caveat that the explanation of this result is speculative and needs further confirmation. In addition, the incidence and severity of AEs after PLD monotherapy were generally consistent with the experience in earlier studies on DTs (20, 27), or other tumor types (28, 29), with no new safety concerns identified. The evaluation per PRO-CTCAE provided insights from the patient's perspective and also supported the favorable tolerability of PLD. Altogether, these results support PLD as an acceptable and safe option for patients with advanced DTs.

Despite positive outcomes, our trial has several limitations. First, the interpretation of our results is limited by a single-center design and relatively small sample size, which had the potential to introduce bias, although these aspects are consistent with other phase III trials in this rare disease. Because of the absence of consensus about the most effective regimen, choosing an active comparator for the randomized trial on DTs remains a challenging problem; therefore, this trial used a placebo control with similarity to previous randomized trials in this setting. Additionally, our study included a high proportion of patients with multifocal tumors but observed a low incidence of familial adenomatous polyposis, which is not entirely consistent with existing epidemiology. We hypothesized that the complex treatment histories, the extended duration of therapy, and the tumor's invasion into multiple anatomic sites contribute to

the formation of discontinuous masses. Incomplete data for PROs are another frequent issue, warranting cautious interpretation of results derived from the PRO-CTCAE assessment. The limited number of pediatric patients restricted the generalizability of our findings, but the encouraging efficacy and favorable safety profile of PLD suggest that it may present a promising treatment option for this population. Lastly, the current follow-up period may not be sufficient to adequately assess the cardiotoxicity associated with low cumulative doses of PLD. Follow-up remains ongoing and will contribute to a more comprehensive understanding of the crossover outcomes, disease progression, cardiotoxicity, female fertility preservation, and quality of life.

In conclusion, this was a positive trial that met its stated goals, demonstrating efficacy in the same range as other treatments previously assessed in randomized trials, such as methotrexate-vinblastine, sorafenib, pazopanib, or niraparicestat, with a favorable safety profile. Therefore, PLD could be added to the available options to be discussed with patients with progressive or symptomatic DTs according to their single and unique situation.

Data Availability

To advance science and medicine while protecting patient privacy, deidentified individual participant data, together with a data dictionary defining each field in

the dataset, will be made available to other researchers upon request to the corresponding author. The study protocol and statistical analysis method are provided in the appendix of this publication.

Authors' Disclosures

No disclosures were reported.

Authors' Contributions

H. Xu: Conceptualization, resources, data curation, methodology, writing—original draft. **J. Hu:** Data curation, formal analysis, writing—original draft. **Yu Zhang:** Resources, writing—original draft. **A. Zhuang:** Resources. **Yingchun Zhang:** Resources. **J. Li:** Resources. **H. Wu:** Resources. **G. Song:** Resources. **Chuanquan Deng:** Resources. **H. Chen:** Resources. **W. Chen:** Resources. **Chuangzhong Deng:** Resources. **A. Wang:** Resources. **H. Feng:** Resources.

J. Lu: Resources. **Q. Tang:** Conceptualization, resources, supervision. **X. Zhu:** Conceptualization, resources, supervision. **J. Wang:** Conceptualization, data curation, supervision, investigation, methodology.

Acknowledgments

We would like to thank the patients who participated in this study and their families and the study teams at each participating institution.

Note

Supplementary data for this article are available at Clinical Cancer Research Online (<http://clincancerres.aacrjournals.org/>).

Received August 26, 2025; revised October 27, 2025; accepted January 5, 2026; posted first January 8, 2026.

References

- Penel N, Coindre J-M, Bonvalot S, Italiano A, Neuville A, Le Cesne A, et al. Management of desmoid tumours: a nationwide survey of labelled reference centre networks in France. *Eur J Cancer* 2016;58:90–6.
- Kasper B, Baumgarten C, García J, Bonvalot S, Haas R, Haller F, et al. An update on the management of sporadic desmoid-type fibromatosis: a European consensus initiative between sarcoma PATients EuroNet (SPAEN) and European Organization for Research and Treatment of Cancer (EORTC)/Soft Tissue and Bone Sarcoma Group (STBSG). *Ann Oncol* 2017;28:2399–408.
- Bektas M, Bell T, Khan S, Tumminello B, Fernandez MM, Heyes C, et al. Desmoid tumors: a comprehensive review. *Adv Ther* 2023;40:3697–722.
- Riedel RF, Agulnik M. Evolving strategies for management of desmoid tumor. *Cancer* 2022;128:3027–40.
- Tsagozis P, Stevenson JD, Grimer R, Carter S. Outcome of surgery for primary and recurrent desmoid-type fibromatosis. A retrospective case series of 174 patients. *Ann Med Surg (Lond)* 2017;17:14–9.
- Cuomo P, Scoccianti G, Schiavo A, Tortolini V, Wigley C, Muratori F, et al. Extra-abdominal desmoid tumor fibromatosis: a multicenter EMSOS study. *BMC Cancer* 2021;21:437.
- Desmoid Tumor Working Group. The management of desmoid tumours: a joint global consensus-based guideline approach for adult and paediatric patients. *Eur J Cancer* 2020;127:96–107.
- Kasper B, Baldini EH, Bonvalot S, Callegaro D, Cardona K, Colombo C, et al. Current management of desmoid tumors: a review. *JAMA Oncol* 2024;10:1121–8.
- Gounder MM, Mahoney MR, Van Tine BA, Ravi V, Attia S, Deshpande HA, et al. Sorafenib for advanced and refractory desmoid tumors. *Engl J Med* 2018;379:2417–28.
- Toulmonde M, Pulido M, Ray-Coquard I, Andre T, Isambert N, Chevreau C, et al. Pazopanib or methotrexate–vinblastine combination chemotherapy in adult patients with progressive desmoid tumours (DESMOPAZ): a non-comparative, randomised, open-label, multicentre, phase 2 study. *Lancet Oncol* 2019;20:1263–72.
- Gounder M, Ratan R, Alcindor T, Schöffski P, Van Der Graaf WT, Wilky BA, et al. Nirogacestat, a γ -secretase inhibitor for desmoid tumors. *Engl J Med* 2023;388:898–912.
- Azzarelli A, Gronchi A, Bertulli R, Tesoro Tess JD, Baratti D, Pennacchioli E, et al. Low-dose chemotherapy with methotrexate and vinblastine for patients with advanced aggressive fibromatosis. *Cancer* 2001;92:1259–64.
- Skapek SX, Ferguson WS, Granowetter L, Devidas M, Perez-Atayde AR, Dehner LP, et al. Vinblastine and methotrexate for desmoid fibromatosis in children: results of a Pediatric Oncology Group Phase II Trial. *J Clin Oncol* 2007;25:501–6.
- Garbay D, Le Cesne A, Penel N, Chevreau C, Marec-Berard P, Blay J-Y, et al. Chemotherapy in patients with desmoid tumors: a study from the French Sarcoma Group (FSG). *Ann Oncol* 2012;23:182–6.
- Liu X, Wang H, Wu X, Hong X, Luo Z. Phase II study of doxorubicin and thalidomide in patients with refractory aggressive fibromatosis. *Invest New Drugs* 2018;36:114–20.
- Nathenson MJ, Hu J, Ratan R, Somaiah N, Hsu R, DeMaria PJ, et al. Systemic chemotherapies retain antitumor activity in desmoid tumors independent of specific mutations in CTNNB1 or APC: a multi-institutional retrospective study. *Clin Cancer Res* 2022;28:4092–104.
- Tahover E, Patil YP, Gabizon AA. Emerging delivery systems to reduce doxorubicin cardiotoxicity and improve therapeutic index: focus on liposomes. *Anticancer Drugs* 2015;26:241–58.
- Theodoulou M, Hudis C. Cardiac profiles of liposomal anthracyclines: greater cardiac safety versus conventional doxorubicin? *Cancer* 2004;100:2052–63.
- Constantinidou A, Jones RL, Scurr M, Al-Muderis O, Judson I. Advanced aggressive fibromatosis: effective palliation with chemotherapy. *Acta Oncologica* 2011;50:455–61.
- Constantinidou A, Jones RL, Scurr M, Al-Muderis O, Judson I. Pegylated liposomal doxorubicin, an effective, well-tolerated treatment for refractory aggressive fibromatosis. *Eur J Cancer* 2009;45:2930–4.
- Cassidy MR, Lefkowitz RA, Long N, Qin L-X, Kirane A, Sbaity E, et al. Association of MRI T2 signal intensity with desmoid tumor progression during active observation: a retrospective cohort study. *Ann Surg* 2020;271:748–55.
- Penel N, Le Cesne A, Bonvalot S, Giraud A, Bompas E, Rios M, et al. Surgical versus non-surgical approach in primary desmoid-type fibromatosis patients: a nationwide prospective cohort from the French Sarcoma Group. *Eur J Cancer* 2017;83:125–31.
- He XD, Zhang YB, Wang L, Tian ML, Liu W, Qu Q, et al. Prognostic factors for the recurrence of sporadic desmoid-type fibromatosis after macroscopically complete resection: analysis of 114 patients at a single institution. *Eur J Surg Oncol* 2015;41:1013–9.
- Napolitano A, Mazzocca A, Spalato Ceruso M, Minelli A, Baldo F, Badalamenti G, et al. Recent advances in desmoid tumor therapy. *Cancers (Basel)* 2020;12:2135.
- Blank N, Laskov I, Kessous R, Kogan L, Lau S, Sebag I, et al. Absence of cardiotoxicity with prolonged treatment and large accumulating doses of pegylated liposomal doxorubicin. *Cancer Chemother Pharmacol* 2017;80:737–43.
- Li X-R, Cheng X-H, Zhang G-N, Wang X-X, Huang J-M. Cardiac safety analysis of first-line chemotherapy drug pegylated liposomal doxorubicin in ovarian cancer. *J Ovarian Res* 2022;15:96.
- Shimizu K, Kawashima H, Kawai A, Yoshida M, Nishida Y. Effectiveness of doxorubicin-based and liposomal doxorubicin chemotherapies for patients with extra-abdominal desmoid-type fibromatosis: a systematic review. *Jpn J Clin Oncol* 2020;50:1274–81.
- Jiang H, Li H, Song G, Di L, Shao B, Yan Y, et al. Pegylated liposomal doxorubicin (Duomeisu®) monotherapy in patients with HER2-negative metastatic breast cancer heavily pretreated with anthracycline and taxanes: a single-arm, phase II study. *Breast Cancer Res Treat* 2023;199:67–79.
- Wang T, Tang J, Yang H, Yin R, Zhang J, Zhou Q, et al. Effect of apatinib plus pegylated liposomal doxorubicin vs pegylated liposomal doxorubicin alone on platinum-resistant recurrent ovarian cancer: the APPROVE randomized clinical trial. *JAMA Oncol* 2022;8:1169–76.